

## Technology Stack

|                      |                          |
|----------------------|--------------------------|
| <b>Date</b>          | 29 September 2026        |
| <b>Team ID</b>       | 6ab4cf016603573ccfb54f03 |
| <b>Project Name</b>  | AI-WeatherWise           |
| <b>Maximum Marks</b> | 2 Marks                  |

### Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen   | Justification / Purpose                                                                                                                 |
|------|--------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | Node.js, Express.js | Used to build RESTful APIs, handle client requests, process weather data, and manage communication with external APIs and the database. |

|   |                                     |                     |                                                                                                                     |
|---|-------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------|
| 2 | <b>Backend / Server-Side</b>        | Node.js, Express.js | Used to build RESTful APIs, handle client requests, process weather data, and manage API communication.             |
| 3 | <b>Database / Data Storage</b>      | MongoDB             | Used to store weather-related data in a flexible and scalable NoSQL database.                                       |
| 4 | <b>Cloud / Hosting / Deployment</b> | Render              | Used to deploy and host the backend API with easy configuration and cost-effective deployment.                      |
| 5 | <b>Version Control &amp; CI/CD</b>  | GitHub              | Used for version control, source-code management, collaboration, and maintaining different versions of the project. |

|          |                                           |             |                                                                                                                                    |
|----------|-------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------|
|          |                                           |             |                                                                                                                                    |
| <b>6</b> | <b>Third-Party APIs /<br/>Other Tools</b> | Weather API | Used to fetch real-time weather data such as temperature, humidity, wind speed, and weather conditions for the requested location. |